accounts

opening a file?

say, private file on portal

for example:

```
open("/u/bradjc/private.txt", O_RDONLY)
```

on Linux: makes system call

kernel needs to decide if this should work or not

system call = ???

(more detail in later lecture)

way for programs to make request to OS:

1. program sets registers to indicate requested operation
   - example: “open foo.txt”
     - register 1: “system call number” (eg, open)
     - register 2: “arguments” (eg, foo.txt)
2. program runs special instruction (x86-64: `syscall`)
3. hardware switches into *privileged* mode + runs OS function
4. OS function decodes requested operation from registers
5. OS function *decides* if operation is allowed to proceed

deciding whether syscall should proceed?

```
open("/u/bradjc/private.txt", O_RDONLY)
```

argument: needs extra metadata

what would be wrong with using:

- system call arguments?
- where the code calling open came from?

user IDs

most common way OSes identify "who" process belongs to:

- process = instance of running program (w/ own registers+memory)
- (we'll be more specific about processes later)

(unspecified for now) procedure sets user IDs

every process has a user ID

user ID used to decide what process is authorized to do

POSIX user IDs

```
uid_t geteuid(); // get current process's "effective" user ID
```

process's user identified with unique number

core part of OS only knows number (not name!)

- core, always loaded part of OS = "kernel"
- the part of the OS with extra privileges with hardware
- the part of the OS that enforces program restrictions

effective user ID is used for all permission checks

standard programs/library maintain number to name mapping

- `/etc/passwd` on typical single-user systems
- network database on department machines

POSIX groups

```
// process's "effective" group ID
gid_t getegid(void);

// process's extra group IDs
int getgroups(int size, gid_t list[]);
```

POSIX also has *group IDs*

like user IDs: kernel (= core part of OS) only knows numbers

- standard library+databases for mapping to names

also process has some other group IDs — we'll talk later

id command

```
: /u/bjc8c ; id
uid=858545(bjc8c) gid=90002(csfaculty)
groups=90002(csfaculty),150015(slurm-cs-brad-campbell)
```

id command displays uid, gid, group list

names looked up in database

- kernel doesn’t know about this database
- code in the C standard library

groups that don't correspond to users

example: video group for access to monitor

put process in video group when logged in directly

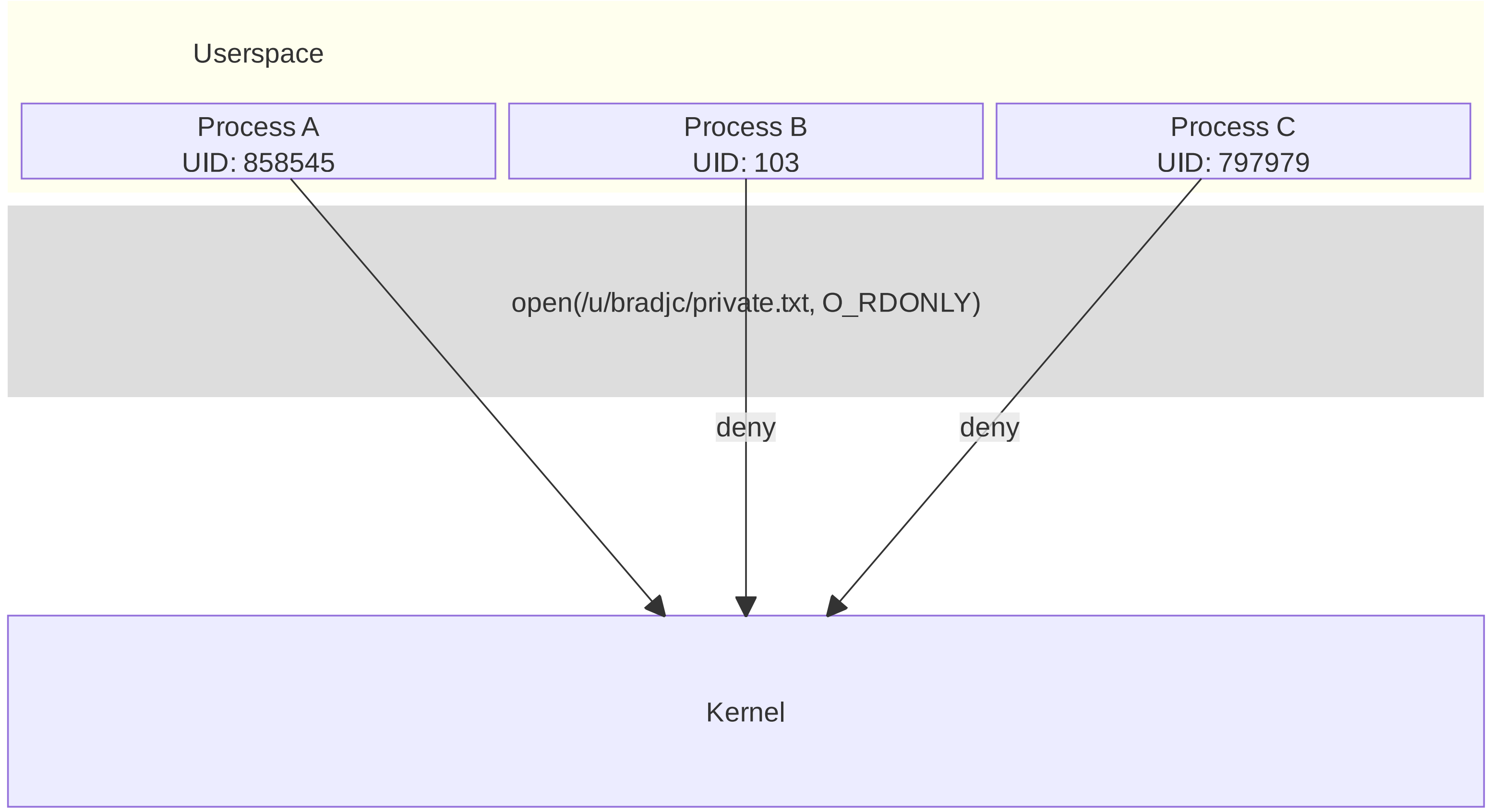
don't do it when SSH'd in

user id and group id establish *identity*

who is running a process or accessing a resource

but also need *permissions*

- what users can access



POSIX file permissions

POSIX files support a *per-file* access control list

- interface is very narrow/limited
- binary permissions are: **read**, **write**, **execute**
- set for three categories: **user**, **group**, **other**

one user ID + read/write/execute bits for user

- “owner” — also can change permissions

one group ID + read/write/execute bits for group

default setting — read/write/execute

on directories, “execute” means “search” instead

permissions encoding

permissions encoded as 9-bit number, can write as octal: `XYZ`

octal divides into three 3-bit parts:

- user permissions (X), group permissions (Y), other permission (Z)

each 3-bit part has a bit for ‘read’ (4), ‘write’ (2), ‘execute’ (1)

- binary number: `0bRWE`

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`700` — user read+write+execute; group none; other none

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`451`?

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- binary number: `0bRWE`

`700` — user read+write+execute; group none; other none

`451`?

`451` — user read; group read+execute; other execute

chmod — exact permissions

```
chmod 700 file
chmod u=rwx,go= file
```

user read write execute; group/others no accesss

```
chmod 451 file
chmod u=r,g=rx,o=x file
```

user read; group read/execute; others execute

chmod — adjusting permissions

```
chmod u+rx foo
```

add user read and execute permissions

leave other settings unchanged

```
chmod o-rwx,u=rx foo
```

remove other read/write/execute permissions
set user permissions to read/execute
leave group settings unchanged

POSIX/NTFS ACLs

more flexible access control lists

list of (user or group, read or write or execute or …)

supported by NTFS (Windows)

a version standardized by POSIX, but usually not supported

POSIX ACL syntax

```
# group students have read+execute permissions
group:students:r-x
# group faculty has read/write/execute permissions
group:faculty:rwx
# user mst3k has read/write/execute permissions
user:mst3k:rwx
# user tj1a has no permissions
user:tj1a:---

# POSIX acl rule:
    # user take precedence over group entries
```

POSIX ACLs on command line

```
getfacl file
```

add line to ACL:

```
setfacl -m 'user:tj1a:---' file
```

REMOVE line from acl:

```
setfacl -x 'user:tj1a' file
```

add to acl, but read what to add from a file:

```
setfacl -M acl.txt file
```

remove from acl, but read what to remove from a file:

```
setfacl -X acl.txt file
```

ACLs establish permissions

identity: *who* is running a process or accessing a resource

permissions: who has *what access* to a file or resource

now need *enforcement*

- ensure that permissions are implemented correctly

authorization checking on Unix

request made to core part of OS = system call

handler for system calls checks permissions

- no relying on libraries, etc. to do checks

file operations (eg: open, rename, …)

- file/directory permissions include UID or GID

process operations (eg: kill, …)

- process UID = user UID

…

superuser

user ID `0` is special

superuser or *root*

- (non-Unix) or Administrator or SYSTEM or …

some OS functionality: only works for uid `0`

- shutdown, mount new file systems, etc.

automatically passes all (or almost all) permission checks

superuser v kernel mode

processor has two modes

- kernel mode (what core part of OS uses)
- user mode (every thing else)

programs running as *superuser still in user mode*

- just change in how OS acts when program asks for things

superuser : OS :: kernel mode : hardware

certain hardware instructions/operations only enabled in kernel mode

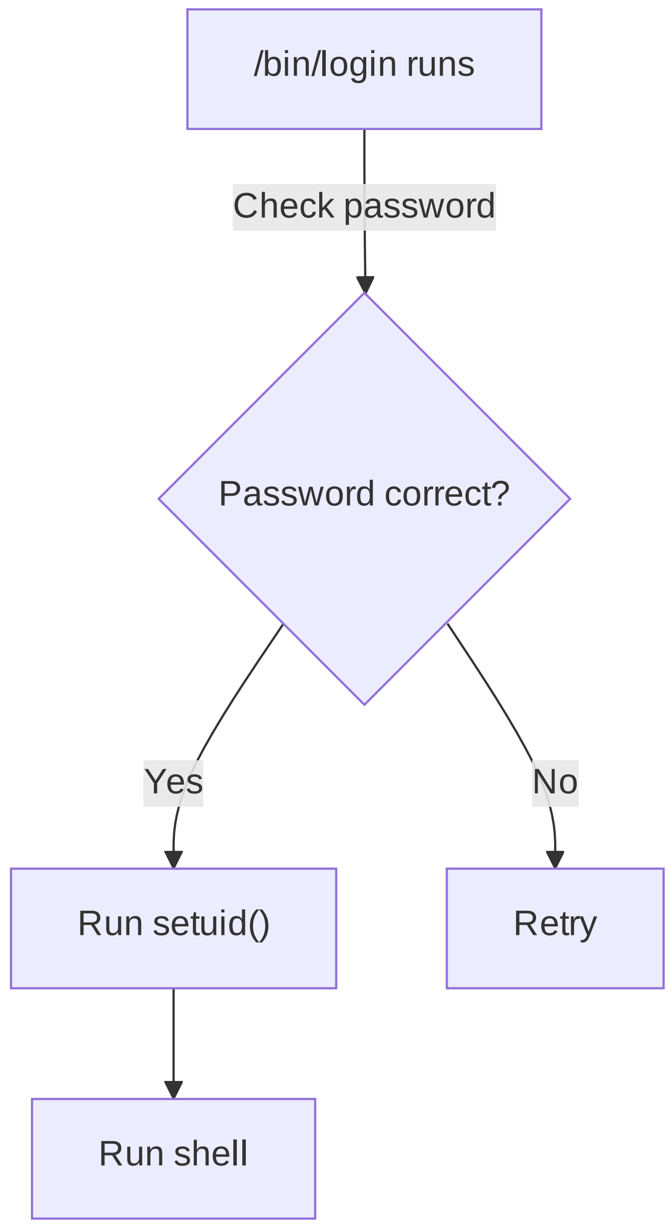
how does login work?

```
somemachine login: jo
password: ********

jo@somemachine$ ls
...
```

this is a program which…

1. checks if the password is correct, and
2. changes user IDs, and
3. runs a shell



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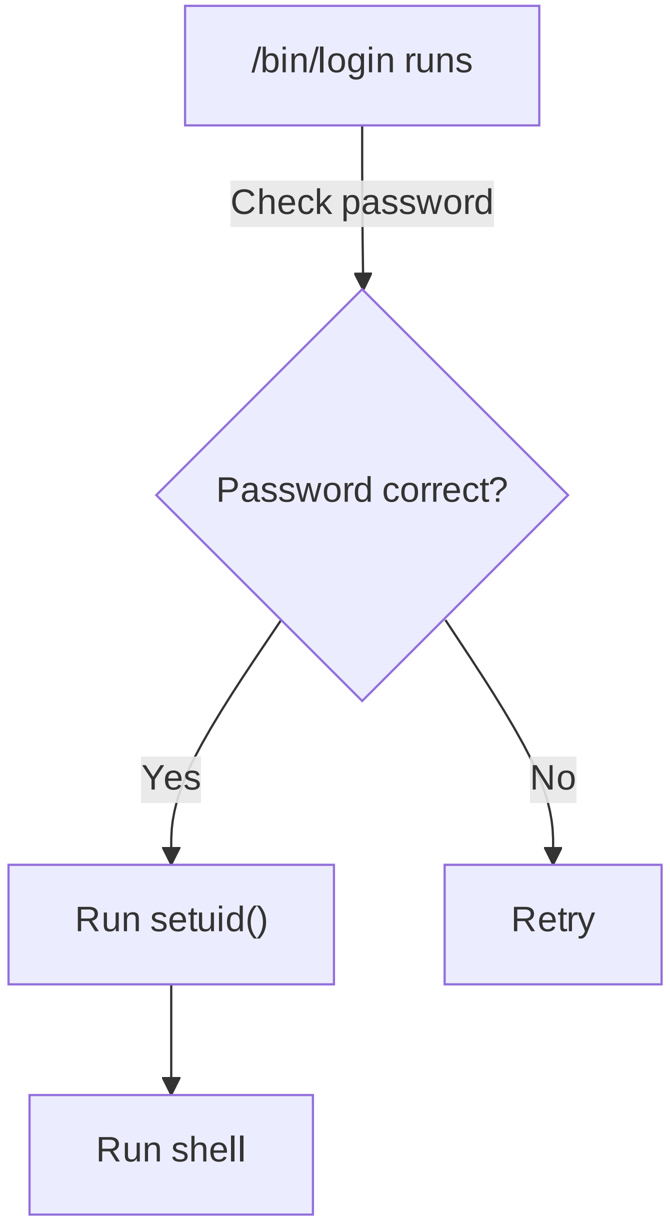

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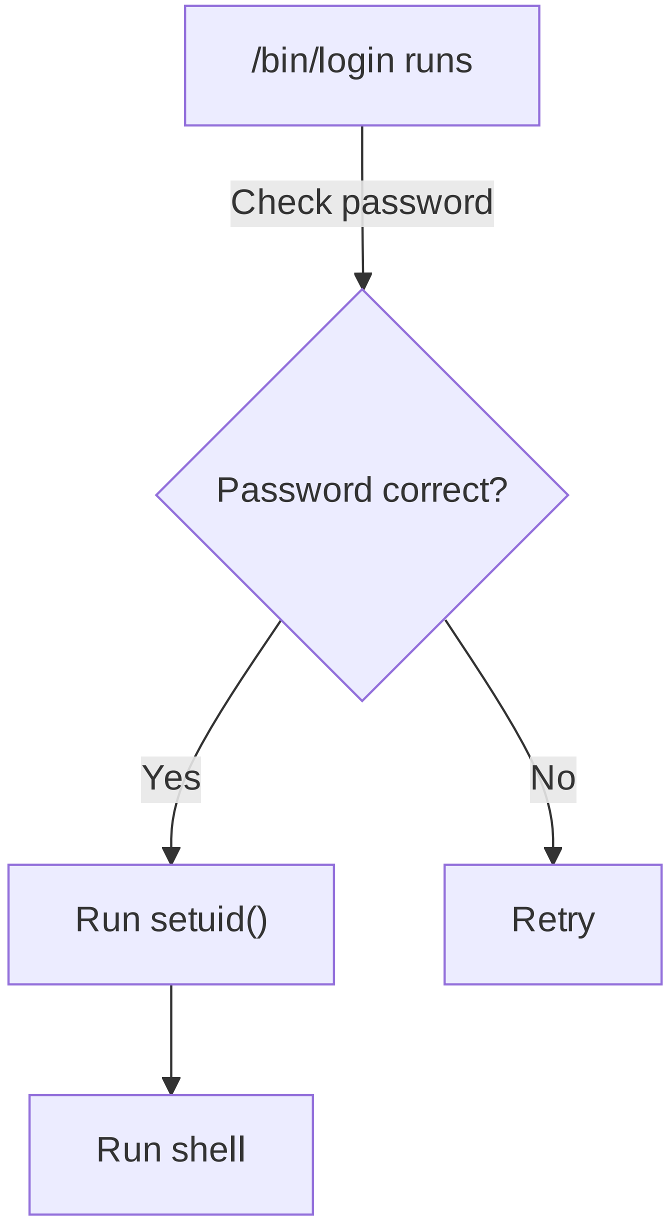


Unix password storage

typical single-user system: `/etc/shadow`

- only readable by root/superuser

department machines: network service

- Kerberos / Active Directory:
- server takes (encrypted) passwords
- server gives tokens: "yes, really this user"
- can cryptographically verify tokens come from server

aside: beyond passwords

/bin/login entirely user-space code

only thing special about it: when it's run

could use any criteria to decide, not just passwords

- physical tokens
- biometrics
- ...

changing user IDs

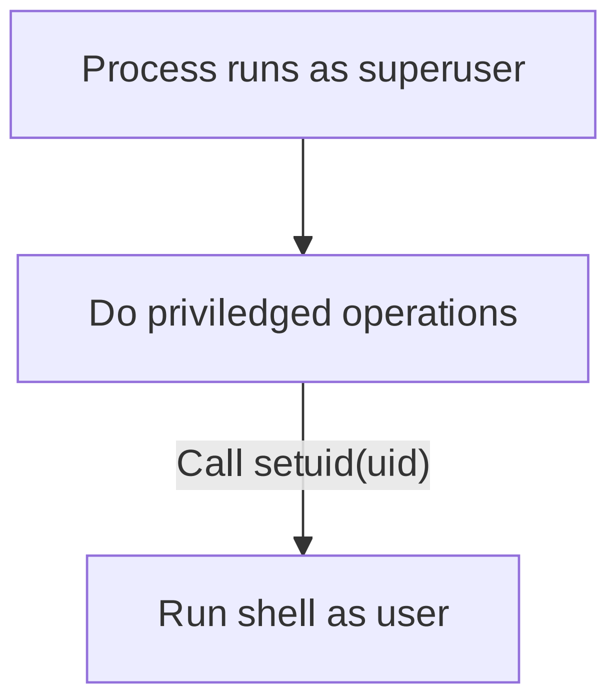
```
int setuid(uid_t uid);
```

if superuser: sets effective user ID to arbitrary value

- and a "real user ID" and a "saved set-user-ID" (we'll talk later)

system starts as superuser and login programs run as superuser

- voluntarily restrict own access before running shell, etc.



sudo

```
tj1a@somemachine$ sudo restart
Password: *********
```

sudo: run command with superuser permissions

- process started by non-superuser

issue: what makes the `sudo` command privileged?

- recall: normal command inherits UID from user who runs it (ie, non-superuser UID)
- can't just call `setuid(0)`

set-user-ID sudo

extra metadata bit on executables: set-user-ID

if set: `exec()` syscall changes effective user ID to *owner's* ID

- "extra" user IDs track what original user was

`sudo` program: owned by root, marked set-user-ID

- sudo's code: if (original user allowed) …; else print error

marking setuid: `chmod u+s`

uses for setuid programs

mount USB stick

- setuid program controls option to kernel mount syscall
- make sure user can’t replace sensitive directories
- make sure user can’t mess up filesystems on normal hard disks
- make sure user can’t mount new setuid root files

control access to device — printer, monitor, etc.

- setuid program talks to device + decides who can

write to secure log file

- setuid program ensures that log is append-only for normal users

bind to a particular port number <1024

- setuid program creates socket, then becomes not root

set-user ID programs are very hard to write

what if stdin, stdout, stderr start closed?

what if the PATH env. var. set to directory of malicious programs?

what if `argc == 0`?

what if dynamic linker env. vars are set?

what if some bug allows memory corruption?

…

privilege escalation

privilege escalation — vulnerabilities that allow more privileges

code execution/corruption in utilities that run with high privilege

- e.g. buffer overflow, command injection
- login, sudo, system services, …
- bugs in system call implementations

logic errors in checking delegated operations